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Class: BSCS/BSSE

Assignment No. 1

Important Notes:

Assignments will be submitted on time. You will have 10 hours to submit the assignment after the due date & time. However, 10% marks will be deducted for every passing hour after the due time. Copying/Cheating whole or part of the assignment from anywhere without proper credit/references will not be tolerated. Whether you have copied or your work has been copied by someone else, you will get zero marks in that assignment.

Exercise 1: Write C# code to declare a variable to store the age of a person. Then the output of the program is as an example shown below:

You are 20 years old.

Exercise 2: Write C# code to display the asterisk pattern as shown below:

```
*****
*****
*****
*****
*****
```

Exercise 3: Write C# code to declare two integer variables, one float variable, and one string variable and assign 10, 12.5, and "C# programming" to them respectively. Then display their values on the screen.

Exercise 4: Write C# code to prompt a user to input his/her name and then the output will be shown as an example below:

Hello John!

Exercise 5: Write a C# program to detect key presses.

If the user pressed number keys (from 0 to 9), the program will display the number that is pressed, otherwise the program will show "Not allowed".

Exercise 6: Write a C# program that allows the user to choose the correct answer of a question.

See the example below:

What is the correct way to declare a variable to store an integer value in C#?

- a. int 1x=10;
- b. int x=10;
- c. float x=10.0f;
- d. string x="10";

Choose the answer letter: c

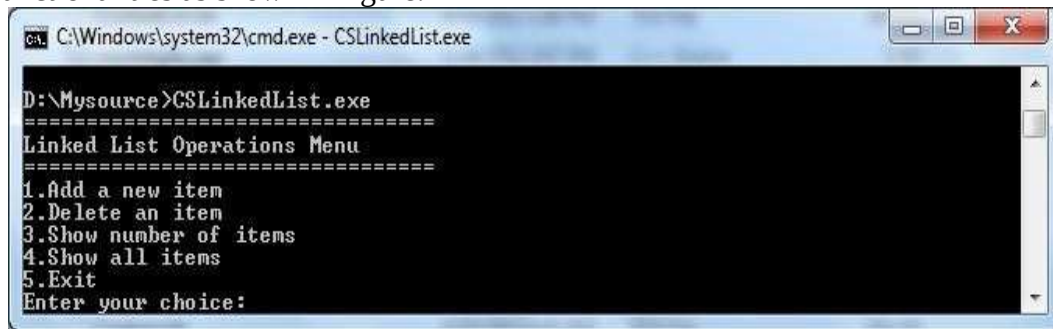
Incorrect choice!

Exercise 7: Write C# code to print the following pattern:

```
1*****
12*****
123*****
1234***
12345**
123456*
1234567
```

Exercise 8: By using the sequential search algorithm, write C# code to search for an element of an integer array of 10 elements.

Exercise 9: By using C# objected-oriented concept, create a list using GENERICS. Provide all functionalities as shown in figure.



```
C:\Windows\system32\cmd.exe - CSLinkedList.exe

D:\Mysource>CSLinkedList.exe
=====
Linked List Operations Menu
=====
1.Add a new item
2.Delete an item
3.Show number of items
4.Show all items
5.Exit
Enter your choice:
```